A single renormalized energy denominator controls the nonperturbative loop expansion of quantum spin ice

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Abstract

The low-energy theory of quantum spin ice is a loop expansion: an alternating flip of L spins around a closed path enters the ice-manifold Hamiltonian with amplitude $K_L^{(0)} = 2\binom{L-2}{L/2-1}|J_\pm|^{L/2}/J_{zz}^{L/2-1}$, built from $L/2-1$ energy denominators each taken to be J_{zz}. We measure these amplitudes nonperturbatively, on finite clusters from which the noncontractible torus processes have been removed exactly, and find that the entire nonperturbative content is a single number. Writing $\mu \equiv J_{zz}/D_{\rm eff}$, the measured amplitudes obey $K_L = K_L^{(0)}\mu^{L/2-1}$ for $L = 8, 10, 12$ with a spread of 1–3.5% across the whole coupling range and two virtual depths: *every* denominator is renormalized from J_{zz} to a common $D_{\rm eff}(J_\pm)$. Two exact sum rules make $D_{\rm eff}$ well defined rather than fitted — the second-order amplitude vanishes identically and the first spectral moment is exactly $12|J_\pm|^3$ — so that $D_{\rm eff} = \sqrt{M_1/K_6}$ equals J_{zz} in leading perturbation theory by construction. We measure $D_{\rm eff}/J_{zz} = 1.17 \to 3.35$ over $|J_\pm|/J_{zz} = 0.03 \to 0.30$, and verify that it is intensive: doubling the cluster from 16 to 32 sites moves it by less than 10%. The hexagon amplitude is therefore suppressed by μ^2, a factor of 11 at $|J_\pm|/J_{zz} = 0.30$ and 4.3 at the couplings relevant to the cerium pyrochlores. Three consequences follow. The emergent photon scale is several times colder than the standard estimate. The two peaks of the specific heat do not merge: the perturbative ratio grows as $|J_\pm|^3$ but μ^2 falls faster, and the merging seen both in perturbation theory and in naive periodic exact diagonalization is an artifact. For $Ce_2Hf_2O_7$ and $Ce_2Zr_2O_7$ the gauge feature lies at a few mK, below every temperature yet measured, so a single broad hump is the expected signature rather than evidence for large transverse exchange. We identify the entropy released below the observed hump, which must be $\frac{R}{2}\ln\frac{3}{2}$, as the parameter-free measurement that decides between these readings.

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I. INTRODUCTION

Quantum spin ice (QSI) is the three-dimensional compact $U(1)$ lattice gauge theory realized by a spin-1/2 pyrochlore magnet with dominant Ising exchange [1–4]. Its standard derivation restricts the Hilbert space to the two-in–two-out ice manifold and generates, at third order in the transverse exchange, a ring-exchange term around each hexagon. Every quantitative statement about the emergent photon — its speed, its bandwidth, the temperature of the thermodynamic crossover into the gauge sector — is anchored on the coefficient of that term,

$$g_6 = \frac{12|J_\pm|^3}{J_{zz}^2}. \tag{1}$$

The derivation is controlled at small $|J_\pm|/J_{zz}$. The materials are not: the cerium pyrochlores sit near $|J_\pm|/J_{zz} \simeq 0.13$–$0.16$, where the third-order truncation is an assumption rather than a limit.

This paper measures the loop amplitudes directly instead of assuming them. The measurement is possible because two obstructions have been removed. The first is topological: a periodic cluster is a torus, and supports ice-preserving spin rearrangements that close only through a periodic image, entering at order $J_\pm^2$ — one order *before* the physical hexagon — and manufacturing the dominant low-energy scale [5]. We work throughout with the transport-masked map of Ref. [5], in which those processes are deleted exactly. The second is that a nonperturbative amplitude must be compared with perturbation theory on equal terms; Sec. IV establishes two exact sum rules that fix the comparison with no fitted quantity.

The result is that the nonperturbative loop expansion has the same form as the perturbative one, with one number changed. Section III reports the collapse; Sec. IV makes it exact; Sec. V asks where the renormalization comes from; Sec. VI works out what it does to the emergent photon, to the two-peak specific heat, and to the two materials in which these questions are currently being asked.

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II. MODEL, CLUSTER AND MASK

We study the nearest-neighbour XXZ model on the pyrochlore lattice,

$$H = H_0 + V, \qquad H_0 = J_{zz} \sum_t \tfrac{1}{2} Q_t^2, \qquad V = -J_\pm \sum_{\langle ij \rangle} \left(S_i^+ S_j^- + \text{h.c.} \right), \tag{2}$$

with $J_{zz} > 0$ and $Q_t = \sum_{i \in t} S_i^z$. P projects onto the ice manifold ($Q_t = 0$ for all t), $Q = 1 - P$. Energies are in units of J_{zz} and $x \equiv |J_\pm|/J_{zz}$.

Two clusters are used. The 16-site conventional cubic cell (cubic-16, 90 ice states) is small enough that the complete virtual space is affordable, so every quantity there is exact. The 32-site face-centred-cubic cluster (FCC-32, 2970 ice states) is the only cluster on which simple loops of perimeter 8, 10 and 12 survive at all: on cubic-16 every such loop wraps the periodic box and is removed by the mask. On FCC-32 the virtual space is truncated to states within ℓ transverse exchanges of the ice manifold (1.4×10^5, 2.5×10^6 and 2.2×10^7 states for $\ell = 1, 2, 3$). Depth is a convergence parameter, unrelated to the loop perimeter L.

The winding-free ice-block map is the transport mask applied to the exact Feshbach map,

$$F(z) = PHP + PVQ\,(z - QHQ)^{-1}\,QVP, \tag{3}$$

with every matrix element connecting different covering-lattice transport sectors deleted [5]. Since PHP is diagonal on the ice manifold, all loop amplitudes live in the second term.

III. THE LOOP AMPLITUDES COLLAPSE ONTO ONE PARAMETER

Each surviving off-diagonal element of the masked map flips the spins on which two ice configurations differ. Following Ref. [5] we retain only *simple* loops — the induced nearest-neighbour subgraph is connected and every vertex has degree exactly two, which excludes products of shorter rings joined at a vertex — and define K_L as the mean modulus of those elements at perimeter L.

Leading perturbation theory gives, for an alternating simple loop of perimeter $L = 2n$ completed by n exchanges,

$$K_L^{(0)} = 2 \binom{L-2}{L/2-1} \frac{|J_\pm|^{L/2}}{J_{zz}^{L/2-1}}, \tag{4}$$

i.e. 12, 40, 140 and 504 for $L = 6, 8, 10, 12$. The structure that matters below is that Eq. (4) contains $n - 1 = L/2 - 1$ energy denominators, each set to J_{zz}.

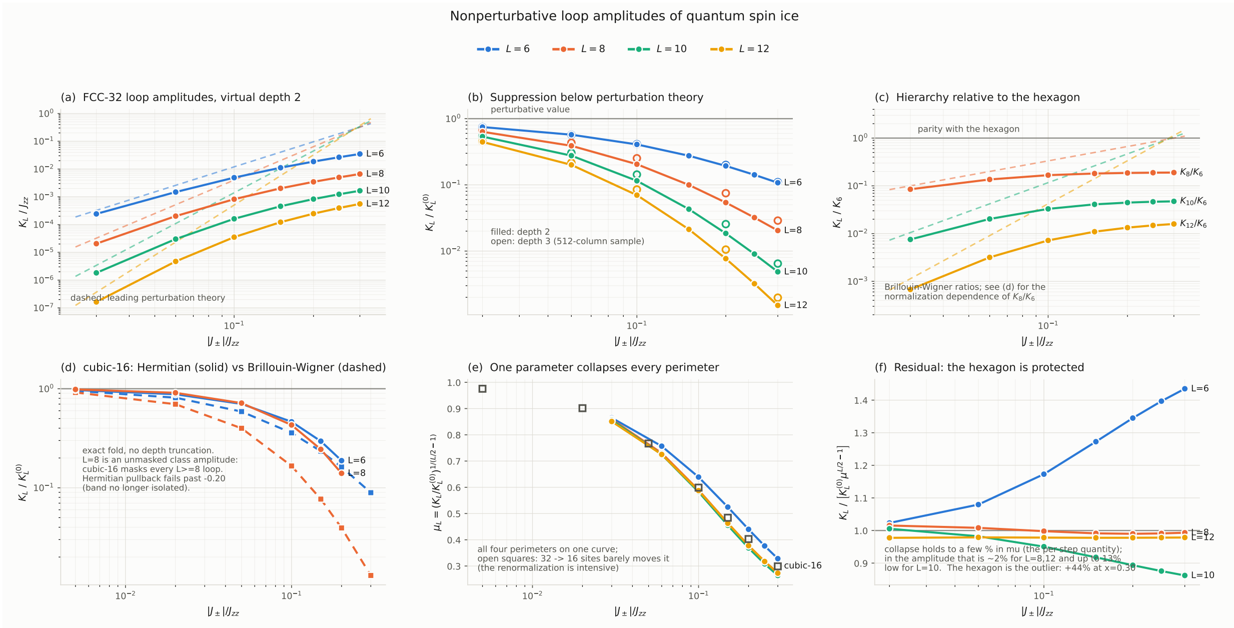

FIG. 1. Nonperturbative loop amplitudes on FCC-32. (a) K_L against coupling with the leading perturbative power laws (dashed); the measured amplitudes fall progressively below them and never cross, although the perturbative curves do near $x = 0.3$. (b) The suppression $K_L/K_L^{(0)}$; open symbols are matched virtual-depth-3 checks and lie systematically above the depth-2 points, so the plotted suppression is a lower bound. (c) The hierarchy relative to the hexagon saturates rather than growing. (d) cubic-16 with the exact fold: the Hermitian (des Cloizeaux) and Brillouin–Wigner normalizations agree at weak coupling and separate at strong coupling, far more for $L = 8$ than for $L = 6$. (e) The collapse: $\mu_L = (K_L/K_L^{(0)})^{1/(L/2-1)}$ for all four perimeters, with cubic-16 as open squares. (f) Residual of the one-parameter law; the hexagon is the single outlier.

We therefore test the hypothesis that the entire nonperturbative effect is a common renormalization μ of each denominator,

$$\boxed{K_L = K_L^{(0)}\, \mu^{L/2-1}} \tag{5}$$

with one μ per coupling and no per-perimeter freedom. Extracting $\mu_L = (K_L/K_L^{(0)})^{1/(L/2-1)}$ separately for each perimeter tests it without fitting anything:

Equation (5) holds to a few percent over the full range, and the same collapse is obtained independently at virtual depth 3. Figure 1(e) shows it directly. The hexagon is the one systematic outlier, lying above the common law by 1% at $x = 0.03$ rising to 21% in μ (equivalently 44% in the amplitude) at $x = 0.30$: the shortest loop is slightly protected relative to the universal law obeyed by the longer ones.

Two immediate corollaries. First, the saturation of the hierarchy in Fig. 1(c) is not new physics: Eq. (5) gives

$$\frac{K_{L+2}}{K_L} = \left(4 - \frac{4}{L}\right) x\,\mu(x), \tag{6}$$

and $x\mu(x)$ saturates ($0.026 \to 0.082$ from $x = 0.03$ to 0.30), so the perturbative linear growth is cancelled by the fall of μ. At $x = 0.30$ this predicts $K_{10}/K_8 = 0.288$ against 0.250 measured. Second, the expansion remains local in perimeter: $K_8/K_6 \to 0.19$, $K_{10}/K_6 \to 0.047$, $K_{12}/K_6 \to 0.016$.

IV. THE RENORMALIZED DENOMINATOR IS EXACTLY DEFINED

Equation (5) invites the reading that each perturbative denominator J_{zz} is replaced by some $D_{\mathrm{eff}} = J_{zz}/\mu$. That reading can be made exact rather than interpretive, using two sum rules.

Decompose the second term of Eq. (3) over the eigenstates of the *full* QHQ, $QHQ|k\rangle = E_k|k\rangle$. For two ice states a, b related by one contractible hexagon, define

$$w_k = \langle b|PVQ|k\rangle\langle k|QVP|a\rangle, \qquad M_n = \sum_k w_k E_k^n = \langle b|VQ(QHQ)^n QV|a\rangle. \tag{7}$$

The amplitude at the physical anchor z_0 (the ground energy of the same truncated microscopic Hamiltonian) is $K_6 = \sum_k w_k/(z_0 - E_k)$. Two facts, both verified numerically to

TABLE I. FCC-32, virtual depth 2. Three independent perimeters spanning two decades in amplitude agree on one number. The $L = 6$ column is shown for contrast and is discussed in the text.

x	μ_6	μ_8	μ_{10}	μ_{12}	spread ($L \geq 8$)
0.03	0.8646	0.8590	0.8559	0.8508	1.0%
0.06	0.7567	0.7302	0.7250	0.7252	0.7%
0.10	0.6390	0.5896	0.5826	0.5874	1.2%
0.15	0.5246	0.4637	0.4552	0.4630	1.8%
0.20	0.4398	0.3780	0.3687	0.3776	2.5%
0.25	0.3765	0.3176	0.3082	0.3172	3.0%
0.30	0.3282	0.2734	0.2640	0.2728	3.5%

machine precision on both clusters and at every coupling:

$$M_0 = \langle b|VQV|a\rangle = 0, \tag{8}$$

$$M_1 = \langle b|VQVQV|a\rangle = 12|J_\pm|^3. \tag{9}$$

Equation (8) is a selection rule: two applications of V move at most four spins, while a hexagon pair differs by six. Equation (9) states that the first spectral moment is exactly the third-order *numerator* of Eq. (1), with no denominators. Note the reorganization relative to textbook perturbation theory: one denominator is explicit in the resolvent, and the second is carried by the eigenvectors of QHQ, which contain V.

Define

$$D_{\rm eff} \equiv \sqrt{M_1/K_6}. \tag{10}$$

This is a definition, but Eqs. (8) and (9) fix its content: in leading perturbation theory $K_6 \to g_6 = 12|J_\pm|^3/J_{zz}^2$ while $M_1 = 12|J_\pm|^3$, so $D_{\rm eff} = J_{zz}$ *exactly.* $D_{\rm eff}$ is therefore the effective value of the product of the two perturbative denominators, and

$$\frac{K_6}{g_6} = \left(\frac{J_{zz}}{D_{\rm eff}}\right)^2 = \mu^2. \tag{11}$$

Because $\sum_k w_k = 0$, the amplitude is not a sum of same-sign contributions: had all intermediate states been degenerate, K_6 would vanish identically. The amplitude is generated by the variation of $(E_k - z_0)^{-1}$ across the intermediate spectrum, and $D_{\rm eff}$ is a weighted mean of $E_k - z_0$, not the energy of any single state.

A. Two controls

Intensivity. $D_{\rm eff}$ is a difference of two absolute energies, so it is an excitation energy and should not scale with volume. Measured on the hexagon channel [Fig. 2(a)], cubic-16 with the exact fold gives $D_{\rm eff}/J_{zz} = 1.374,\ 2.067,\ 3.346$ at $x = 0.06, 0.15, 0.30$, against 1.322, 1.910, 3.058 on FCC-32 at depth 2 — ratios 0.962, 0.924, 0.914. Doubling the number of sites moves $D_{\rm eff}$ by less than 10%. The suppression is a local property of the theory, not a finite-volume or Brillouin–Wigner size-inconsistency artifact.

Normalization. $F(z)$ is a Brillouin–Wigner map: energy dependent, and not a Hermitian effective Hamiltonian. We therefore repeat the hexagon measurement using the Hermitian des Cloizeaux (Löwdin polar) pullback of the exact band onto the ice manifold, which is

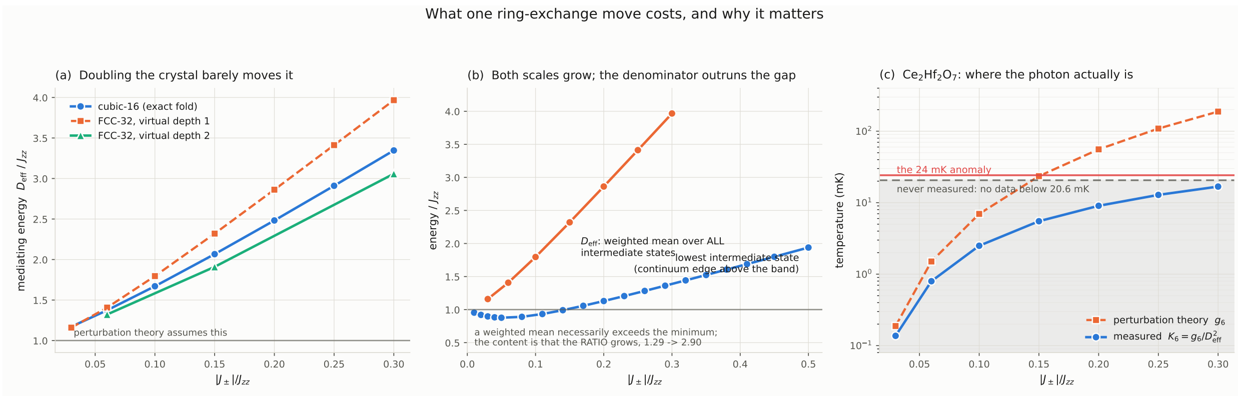

FIG. 2. (a) $D_{\rm eff}$ is intensive: cubic-16 with the complete virtual space and FCC-32 at two virtual depths agree to under 10% at matched depth, against the factor of two an extensive quantity would give. Perturbation theory is the line at 1. (b) $D_{\rm eff}$ against the lowest intermediate-state energy (continuum edge above the ice band). A weighted mean necessarily exceeds a minimum; the content is that the ratio grows, from 1.29 to 2.90. (c) The consequence in physical units for $Ce_2Hf_2O_7$ ($J_{zz} = 0.050$ meV): the perturbative estimate crosses the observed 24 mK anomaly at exactly the fitted couplings, whereas the measured amplitude never reaches the coldest temperature yet measured on the material.

residue free. On cubic-16 the two normalizations agree at weak coupling, $K_6/g_6 = 0.952$ (BW) against 0.971 (Hermitian) at $x = 0.005$, confirming that the comparison with Eq. (1) is properly calibrated, and separate mildly at stronger coupling: 0.234 against 0.296 at $x = 0.15$, and 0.162 against 0.188 at $x = 0.20$. The correction to μ peaks near 1.29 and does *not* track $1/Z$, so the residue is not driving the effect. The correction is, however, strongly perimeter dependent — at $x = 0.20$ it is 1.16 for $L = 6$ but 3.56 for $L = 8$ [Fig. 1(d)] — so the ratios K_L/K_6 are normalization sensitive at the tens-of-percent level even though μ itself is not. Beyond $x \simeq 0.20$ the ice-descended band ceases to be isolated on cubic-16 and the Hermitian object does not exist.

V. WHERE THE RENORMALIZATION COMES FROM

$D_{\rm eff}$ rises from $1.17\,J_{zz}$ to $3.35\,J_{zz}$ across the range studied. Since $D_{\rm eff}$ is a weighted mean of $E_k - z_0$, the growth means that distribution shifts upward. The decomposition is visible in the spectrum itself: between $x = 0.01$ and $x = 0.50$ the bottom of the masked ice band falls from -0.006 to -3.558, whereas the continuum edge falls only from 0.948 to -1.620, so the gap between them grows from 0.954 to 1.939. The ice manifold is stabilized faster than the states above it, and every excitation energy $E_k - z_0$ grows accordingly.

Leading perturbation theory uses unperturbed energies in both places; the exact map uses the true z_0 and the dressed E_k. The difference between the two is the entire effect, and it is the resummation of the higher-order processes that flip the same six spins. We stress what is *not* being claimed: longer loops are distinct operators acting on different spin sets and cannot renormalize K_6. The nontrivial content of Sec. III is that the same μ governs both the correction to the hexagon coefficient and the coefficients of those independent longer-loop operators.

The physical reading — that the ring exchange stabilizes the manifold it acts on and thereby raises its own energy denominator, a self-limiting feedback — is an interpretation of the ordering above, not an independent measurement.

VI. CONSEQUENCES

A. The emergent photon is colder than the standard estimate

The physical ring-exchange coupling is $K_6 = g_6\mu^2$. At the couplings relevant to the cerium pyrochlores this is a suppression of 4.3 ($x = 0.15$), and it reaches 11 at $x = 0.30$. Every quantity anchored on g_6 inherits the factor: the photon bandwidth, the emergent speed of light, the T^3 coefficient of the low-temperature specific heat, and the temperature of the gauge crossover.

B. The two specific-heat peaks do not merge

The standard reading of a single broad hump in a QSI candidate is that the transverse exchange is large enough that the gauge and spinon features have merged. That reading rests on Eq. (1): with the spinon scale $\simeq 0.25J_{zz}$ roughly fixed, the ratio $T_{\mathrm{gauge}}/T_{\mathrm{spinon}} \simeq 48x^3$ reaches unity near $x \simeq 0.28$. Two independent corrections remove the mechanism.

First, the ratio is $48x^3\mu(x)^2$, and since $\mu \simeq (1 + ax)^{-1}$ with $a \approx 8$, the cubic growth that drives the merge becomes *linear* at large x. Merging would require $x \sim a^2/48 \approx 1$, i.e. $|J_\pm| \sim J_{zz}$, far outside any spin ice.

Second, the merging is directly observable as a torus artifact. On cubic-16, comparing the naive periodic full spectrum with the winding-free one at the same coupling:

x	periodic peaks	winding-free peaks
0.20	0.1006, 0.2418	0.0060, 0.0275, 0.348
0.30	0.1634 (one)	0.0056, 0.0419, 0.460
0.45	0.2162 (one)	0.0120, 0.1190, 0.592
0.50	0.2308 (one)	0.0608, 0.1282, 0.632

The periodic calculation merges at $x \geq 0.30$ exactly as the standard picture predicts; the winding-free calculation on the same cluster never does. We note that the periodic two-peak ratio at $x = 0.20$ is 0.416, close to the value measured in $Ce_2Hf_2O_7$ — so a small periodic cluster reproduces the observed ratio through the artifact.

C. The cerium pyrochlores

Taking $J_{zz} = 0.050$ meV = 580 mK, the two estimates of the gauge scale are

x	0.10	0.15	0.20	0.25	0.30
g_6 (mK)	6.96	23.50	55.70	108.8	188.0
K_6 (mK)	2.50	5.50	9.04	12.85	16.79

The perturbative curve crosses the measured 24.2 mK anomaly of $Ce_2Hf_2O_7$ [6] at $x \simeq 0.15$, which is where the material is fitted — so g_6 appears to explain the anomaly. The measured amplitude is several times smaller and does not reach 24 mK anywhere in the range studied; reproducing the anomaly would require $x \approx 0.42$, where the masked hexagon flux on cubic-16 has already fallen to zero and the ice-descended band is no longer isolated.

More usefully, the published data begin at 20.6 mK, and K_6 lies below that floor for every coupling in the plausible range. **The gauge feature has never been inside the measured window.** A single broad hump, as reported for $Ce_2Zr_2O_7$ [7–9], is therefore the expected appearance of a quantum spin ice at accessible temperatures — the observed feature is the spinon and ice-rule crossover, with the gauge feature an order of magnitude colder — and its absence is not evidence against a QSI ground state.

VII. THE MEASUREMENT THAT DECIDES IT

The merged and unmerged readings of a single hump make opposite, parameter-free predictions for the entropy. If the low-temperature feature is the ordering of the ice manifold, it must release the residual ice entropy [10, 11]

$$S_{\text{ice}} = \frac{R}{2} \ln \frac{3}{2} = 1.687 \text{ J mol}^{-1}\text{K}^{-1} = 0.293\, R \ln 2. \tag{12}$$

If the peaks have merged, the observed hump contains both crossovers and carries the full $R \ln 2 = 5.763 \text{ J mol}^{-1}\text{K}^{-1}$. If they have not, the hump carries only $0.71\, R \ln 2 = 4.08 \text{ J mol}^{-1}\text{K}^{-1}$ and Eq. (12) is still missing, waiting at a few mK. No Hamiltonian, fit or cluster enters this test.

Below the gauge crossover the specific heat of two transverse photon polarizations must be $C \propto (k_B T/\hbar c)^3$ [3], with c fixed by K_6 rather than g_6 — a second, independent signature in the same temperature window, and one that distinguishes the photon from a Schottky anomaly, a nuclear contribution or a disorder term.

[PENDING: The entropy integral has not been evaluated on the published $Ce_2Hf_2O_7$ trace. Because the digitized data begin at 20.6 mK, above the low-temperature flank, the unmeasured tail contributes between 0.6 and 1.9 $\text{J mol}^{-1}\text{K}^{-1}$ depending on its exponent, i.e. between 38% and 114% of S_{ice}; the test is therefore likely to be inconclusive without author data below 20 mK, which is itself worth reporting.]

VIII. LIMITATIONS

1. *Virtual depth.* All FCC-32 amplitudes are truncated. The $\ell = 2 \to 3$ change moves μ from 0.274 to 0.293 at $x = 0.30$, so the quoted suppression is a lower bound on μ and the true photon scale is somewhat warmer than plotted. This is far short of the factor needed to place it in the measured window.

2. *Two clusters.* The intensivity test rests on a single doubling, $16 \to 32$ sites.

3. *Normalization.* μ is robust to the Brillouin–Wigner versus Hermitian choice at the 10–30% level, but the ratios K_L/K_6 are not: the correction is 1.16 for $L = 6$ and 3.56

for $L = 8$ at $x = 0.20$. Any statement about the loop hierarchy, as opposed to μ, must quote its normalization.

4. *The hexagon outlier.* $L = 6$ departs from Eq. (5) by up to 44% in amplitude. The collapse is a statement about $L \geq 8$ with the hexagon slightly protected, not a law obeyed by all four perimeters equally.

5. *Absolute temperatures.* Peak positions carry a cluster-dependent combinatorial prefactor (1.070 on cubic-16, 0.521 on FCC-32) that is not converged. The dimensionless statements — the collapse, μ, the peak ratio, the entropy — are the better controlled ones.

IX. CONCLUSION

The nonperturbative loop expansion of quantum spin ice has the same operator content as the perturbative one and differs by a single number. Every energy denominator is renormalized from J_{zz} to a common $D_{\text{eff}}(J_\pm)$, so that $K_L = K_L^{(0)}(J_{zz}/D_{\text{eff}})^{L/2-1}$ to a few percent for $L = 8, 10, 12$; two exact sum rules make D_{eff} a definition that equals J_{zz} in leading perturbation theory, and it is intensive. The hexagon coupling that sets every emergent electrodynamic scale is thereby suppressed by μ^2, a factor of four at material couplings and eleven at the strongest coupling studied.

The consequence for experiment is not a correction but a reinterpretation. The two peaks of the specific heat cannot merge, because the perturbative growth that would merge them is cancelled by the renormalization; the merging seen in perturbation theory and in periodic exact diagonalization is an artifact of both. In the cerium pyrochlores the gauge feature sits at a few mK, below every temperature yet measured, so a single broad hump is what a quantum spin ice should look like, and the feature that has been measured near 24 mK is unlikely to be the emergent photon. The entropy released below the observed hump, which must equal $\frac{R}{2}\ln\frac{3}{2}$ if that hump is the ice crossover, decides between these readings without reference to any model.

ACKNOWLEDGMENTS

The transport-masked construction and the FCC-32 spectra are those of Ref. [5].

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